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"That looks like something I would do": understanding humanities researchers' digital hoarding behaviors in digital scholarship

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Abstract

Purpose – In the era of digital intelligence, individuals are increasingly interacting with digital information in their daily lives and work, and a growing phenomenon known as digital hoarding is becoming more prevalent. Prior research suggests that humanities researchers have unique and longstanding information interaction and management practices in the digital scholarship context. This study therefore aims to understand how digital hoarding manifests in humanities researchers' behavior, identify the influencing factors associated with it, and explore how they perceive and respond to digital hoarding behavior.

Design/methodology/approach – Qualitative research methods enable us to acquire a rich insight and nuanced understanding of digital hoarding practices. In this study, semi-structured interviews were conducted with 20 humanities researchers who were pre-screened for a high propensity for digital hoarding. Thematic analyses were then used to analyze the interview data.

Findings – Three main characteristics of digital hoarding were identified. Further, the research paradigm, digital affordance, and personality traits and habits, collectively influencing the emergence and development of digital hoarding behaviors, were examined. The subtle influence of traditional Chinese culture was encountered. Interestingly, this study found that humanists perceive digital hoarding as a positive expectation (associated with inspiration, aesthetic pursuit, and uncertainty avoidance). Meanwhile, humanists' problematic perception of this behavior is more widely observed — they experience what we conceptualize as an "expectation-perception" gap. Three specific information behaviors related to avoidance were identified as aggravating factors for digital hoarding.

Originality/value – The findings deepen the understanding of digital hoarding behaviors and personal information management among humanities researchers within the LIS field, and implications for humanities researchers, digital scholarship service providers, and digital tool developers are discussed.

Keywords Digital hoarding behavior, Digital scholarship, Personal information management, Digital affordances, Digital humanities

Paper type Research paper

1. Introduction

The omnipresence and increasing efficiency of digital, networked systems or tools available for humanities research, alongside the resulting deluge of digital infrastructure and digital corpora, have together reshaped the current scholarship landscape (Clement and Carter, 2017). This paradigm shift has had a profound impact on how humanities researchers

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interact with digital content and has changed how humanities researchers practice digital curation (Sinn and Soares, 2014). The emergence of digital humanities as a particular focus of humanities over recent years (Given and Willson, 2018; Su *et al.*, 2020) further underscores this shift, bringing about a change in many traditional behaviors, attitudes, understandings, and habits among humanities researchers. Humanities researchers are increasingly engaged in information work within the context of digital data and networked information systems (Clement and Carter, 2017; Sinn and Soares, 2014). They have become "utterly dependent on many different digital tools and resources" (Trace and Karadkar, 2017, p. 505). The ubiquitous availability of digital documents encourages scholars to have a large "digital library" in an online place (Rimmer *et al.*, 2008).

Although the above changes may seem encouraging in general, they have also precipitated problematic or negative responses to personal information management (PIM) from humanities researchers. Recently, several scholars have pointed out that challenges in PIM for humanities researchers' everyday digital scholarly work practices have become more urgent and complex (Alemneh *et al.*, 2005; Poole, 2017). The digital materials archived by humanities researchers are becoming increasingly disorganized, hindering reuse, potentially skewing humanities research (Antonijevic and Cahoy, 2018; Milligan, 2013), and even raising concerns and dilemmas (e.g. information overload, psychological damage, and anxiety) in scholars' information practices (Kamposiori *et al.*, 2019; Krtalić and Dinneen, 2024). Experiments have shown that, when people face information overload, they tend to add more information rather than deleting or reducing it (Adams *et al.*, 2021).

Substantial prior research on humanities researchers' information practices in digital scholarship (summarized below) indicates that the practices and challenges of humanities researchers may be distinctive (Huvila *et al.*, 2022). Stands apart from other disciplines, humanities scholarship has a cumulative nature (Watson-Boone, 1994), their repertories of research material may span even through the researchers' entire academic career (Lin, 2023). And often, researchers use anything as a source of inspiration for their work, including information not directly related to their field (Hellmich and Dinneen, 2023). Humanities researchers are often described as disorganized or overwhelmed, accumulating as much information as possible without proper management. They show reluctance to invest much effort in curation (especially deleting or disposing of), exhibiting a behavior known as "benign neglect" (Bulger *et al.*, 2011). This disordered management or even loss of personal digital records has caused inconvenience in their lives and work (Antonijevic and Cahoy, 2018; Zhao and Yang, 2019). Indeed, without access to the content, the value of information is greatly diminished, rendering digital services useless (Kumpulainen and Late, 2022). All of these behaviors may be related to the potentially prominent issue of *digital hoarding* among humanities researchers, a phenomenon brought about by the digital age, which refers to the constant acquisition, difficulty in discarding, and tendency to clutter digital information items, resulting in stress and disorder (Sedera *et al.*, 2022).

Existing studies examining digital hoarding behaviors have mainly focused on everyday social media settings (Wu *et al.*, 2023) and workplace contexts (McKellar *et al.*, 2023; Sweeten *et al.*, 2018) rather than explicitly addressing digital scholarship contexts. Prior work on digital hoarding has primarily focused on conceptualizing and measuring digital hoarding constructs (Neave *et al.*, 2019; Sedera *et al.*, 2022), potential negative consequences related to digital hoarding (van Bennekom *et al.*, 2015), as well as identifying triggering factors, including utilitarian motives (e.g. evidence and relationship), personal traits (e.g. laziness), and the role of emotional attachments as a major barrier to data deletion (Sweeten *et al.*, 2018). Most studies have focused on the cognitive, psychological, and affective exploration of the individual's influence on digital hoarding behavior.

Only recent studies have examined the influence of organizational culture and policy on digital hoarding (McKellar *et al.*, 2023). Despite the growing recognition of the importance of digital hoarding behavior, relatively little is known about the phenomenon as practiced by humanities researchers in the context of digital academic practices. Understanding how and why digital hoarding behavior occurs, as well as exploring perceptions and coping strategies among those who experience hoarding, is essential for providing insights for various stakeholders, including humanities researchers, information institutions, PIM system designers, and LIS professionals. Building upon the gap identified above, this study aims to explore the following research questions:

RQ1. What characterizes the digital hoarding behavior of humanities researchers in everyday digital scholarship?

RQ2. What factors have triggered digital hoarding behaviors among humanities researchers in their digital scholarship practices?

RQ3. How do humanities researchers perceive and respond to digital hoarding behaviors in their digital scholarly practices?

This study conducted semi-structured interviews with 20 humanities researchers and analyzed the data using thematic analysis. The study of digital hoarding behavior contributes to responding to and expanding the traditional field of LIS research on PIM from an epistemological perspective. It aims to improve the interaction between humanities researchers and their digital information in academic work, facilitating enhanced PIM. From a practical standpoint, the insights derived from our findings can inform the development and optimization of services and tools in humanities, in response to the distinct and genuine needs of humanities researchers in the digital era.

2. Literature review

2.1 Humanities researchers' information behaviors/practices in digital scholarship

With the growing use of technological tools and digital materials for conducting humanities research over the past decade, coupled with the emergence of digital humanities as a distinct focus within humanities scholarship, the working methods of researchers are undergoing significant changes. Humanities scholars' work has traditionally been depicted as centered upon physical substance (Stone, 1982). This includes a longstanding practice among humanities scholars, especially artists, who gather objects both for their creative work and personal enjoyment (Palmer and Neumann, 2002; De la Flor *et al.*, 2010). Today, much of humanities research revolves around the creation of digital objects using digital tools for scholarly works, given the increasing presence of digital information in professional domains (Clement and Carter, 2017; Sinn and Soares, 2014). Infrastructure advancements, such as more affordable storage devices, cloud storage solutions, and the widespread use of social media, play crucial roles in facilitating and advancing this trend, resulting in an increasing material impact (Jung *et al.*, 2010). The theory of digital materiality posits that digital artifacts are considered "material" because they afford or constrain actions (Leonardi, 2010). As digital technology advances, the characteristics of digital objects—namely "Placeless, spaceless, and formless"—become more apparent (Drosopoulou and Cox, 2020). The evolving characteristics of technologies and objects shape how humanities researchers adapt their practices to the new context in beneficial but also problematic ways.

On one hand, humanities scholars increasingly embrace and attach special value to digital video, digital artworks, and expansive archives of digitized manuscripts and documents, etc. They work collaboratively with new tools and technologies, recording and preserving

various digital documents or works in diverse forms (Bulger *et al.*, 2011). Digitization and the increased availability of online information could have an impact not only on scholars' information-seeking and collecting frequently but on other stages of the scholarly workflow as well (Kamposiori *et al.*, 2019; Falciani-White, 2017). Unlike other scholars who have lacked curatorial activity due to their sole focus on the direct use of digital materials (Drosopoulou and Cox, 2020), curatorial activity among humanities researchers has garnered widespread theoretical attention. How to organize and protect these digital materials, and how and where to store them so that they can be used later for their intended purpose become researchers' issues of focus (Krtalić *et al.*, 2016). For example, Trace and Karadkar (2017) examined the capture, organization, and storage of research-related materials (e.g. notes and citations), and highlighted the paradigm shift in the workflow of humanities scholars, moving away from traditional practices of reading and extensive note-taking to more contemporary approaches involving scanning, gathering information, and annotating. Waters (2022) used a six-stage behavior process (i.e. collect, catalog, transcribe, identify, analyze, and publish) to describe key components of a generalized research workflow to accommodate the digital environment through the embrace of digital tools and content.

Nonetheless, on the other hand, infrastructure changes do not always enhance information management and may expose new problems, concerning that although information is more accessible than before, more is stored away and forgotten rather than used (Pontis *et al.*, 2017). Digital materials of all kinds hold unique value for humanities researchers, possibly due to their role in the creative process (Drosopoulou and Cox, 2020), and consequently, they encounter distinctive information management challenges. It has been reported that Humanities researchers may maintain their collections of information over time without specifying their purposes (Post, 2017). The primary concern of most humanities researchers in their information practices was how to manage the information accumulated and created during the research process (Cushing and Dumbleton, 2017), including issues like accessing and reusing data, as well as decisions about what to retain and what to discard time pressure, information overload, and the dispersion of information constitute the main challenges confronting PIM (Chaudhry and Alajmi, 2022). Effectively managing personal information requires maintaining, deleting, adding, and updating the information in these folders through regular reviews (Krtalić and Dinneen, 2024).

Prior work on humanities scholars' PIM and information behavior/practices has begun to indicate that the changing digital academic and technological environment drives them to adapt their information practices to the new context. Humanities scholars, as a specific group, may differ from other groups in terms of saving, organizing, and using activities of digital content, as several pieces of research suggest that humanities researchers always have to archive seemingly relevant information to deal with the vast amount of online information and store it in citation management software, Dropbox, or other storage systems (Antonijevic and Cahoy, 2018). Moreover, they often stray from normative collection procedures and are indifferent to the direct use value and economic value of digital items (Carr, 2024; Hellmich and Dinneen, 2023). As Papson (2014, p. 387) posits, scholars "unsystematically gather materials, never certain when an object might come in handy—like not cleaning a garage full of junk". All of these findings imply that humanities scholars may engage in the potential hoarding of digital materials. Even research suggests that the creative engagement of humanities scholars is closely linked to digital hoarding behavior (Carr, 2024).

2.2 Digital hoarding behavior

The digital age has given rise to a phenomenon of digital hoarding, akin to physical hoarding, which was initially defined in a 2015 case study as an excessive and uncontrolled

accumulation of digital stockpiles that ultimately lead to stress and disruption (van Bennekom *et al.*, 2015). Subsequently, scholars in psychology and information science have confirmed the prevalence of digital hoarding behavior and have measured various facets of this behavior through questionnaires, including the overacquisition of digital content, difficulties in deletion, and digital clutter (Neave *et al.*, 2019; Sedera *et al.*, 2022; Sedera and Lokuge, 2018).

Furthermore, the causes and negative consequences of digital hoarding behaviors in the workplace, knowledge-intensive settings, and social media environments have been extensively discussed. Earlier research on digital hoarding focuses mainly on email hoarding within a workplace setting. Email management is often considered time-consuming and attention-consuming (Dabbish *et al.*, 2005). In the workplace, employees tend to focus on email access and utilization but give little thought to organization and cleanup (McKellar *et al.*, 2020). The organizational culture, responsibilities, and policies in workplaces are distal predictive factors for employee digital hoarding behavior (McKellar *et al.*, 2020, 2023; Sweeten *et al.*, 2018). The proximal factors may be that employees perceive emails as evidence or are unwilling to take the time to delete them (Sweeten *et al.*, 2018), and increasingly large mailboxes make users accustomed to not having to delete emails to save space, creating path dependence on mailbox hoarding (Polites and Karahanna, 2012). The competitive environment, management systems, and employee insecurity within organizations increase the likelihood of digital hoarding or knowledge hoarding among employees in knowledge-intensive organizations (e.g. universities) (ACAR and Seda, 2020; Gormley and Gormley, 2012). Digital hoarding in social media environments such as one-click bookmarks, posts, friends, etc. is primarily influenced by sentimental factors (e.g. emotional attachment, Fear of Missing Out) and instrumental factors (e.g. perceived future usefulness, evidence) (McKellar *et al.*, 2020; Wu *et al.*, 2023). Moreover, digital hoarding can have adverse effects on individuals and organizations, such as stress, financial losses, and cybersecurity issues, and in severe cases, it can be associated with physical hoarding, considered a pathological manifestation of digital hoarding (Sweeten *et al.*, 2018; van Bennekom *et al.*, 2015). There is consensus on the potential consequences of digital hoarding, highlighting that digital hoarding is a common problematic information behavior (Işliyen, 2019).

Prior work on digital hoarding behavior among online user groups has comprehensively addressed digital content, influences, and potential consequences associated with this behavior. However, research on digital hoarding behavior among humanities researchers remains largely unexplored. Conducting in-depth studies of information behavior within these well-defined groups could significantly advance the development of universally applicable behavioral theories (Bawden, 2006; Wilson, 2006). Furthermore, few studies have noted the consideration of researchers' PIM behaviors in broader contexts such as cultural, technological, and social (Drosopoulou and Cox, 2020). Linking digital hoarding behaviors to the social contexts in which humanities researchers are embedded can help explore differences across disciplines and groups more effectively. It is also crucial to acknowledge that managing digital content is highly personal (Dinneen and Frissen, 2020). Therefore, synthesizing the determining factors, detailed and subtle experiences, and responses of humanities researchers is essential for a thorough and detailed exploration of their digital hoarding behaviors.

3. Methods

3.1 Data collection

To understand the phenomenon of digital hoarding behavior among humanist researchers functioning as information workers in today's ever-evolving digital work environment, we conducted semi-structured interviews utilizing a dramaturgical model outlined by Myers and

Newman (2007). Given the challenges of identifying individuals' digital hoarding behavior and the processes and experiences that lead to it through experiments or questionnaire surveys, employing an interview-based, user-experience-centered qualitative approach is appropriate for exploring this theme in research. After obtaining IRB approval from our institution, participants were recruited through interpersonal networks, snowball sampling, and social media platforms, and a pre-interview screening of participants was carried out using Neave *et al.*'s (2019) digital hoarding questionnaire (DHQ). This questionnaire included ten measurement items covering two dimensions—difficulty deleting and accumulating—to comprehensively assess the tendency to digital hoarding, each evaluated using a 7-point Likert scale. Specifically, the dimension of difficulty deleting primarily assesses individuals' resistance to deleting useless data files, while the dimension of continuous accumulation primarily evaluates the tendency of individuals to accumulate data files without effective management (see the appendix for DHQ). High scores were defined as ratings of 5, 6, or 7 for each item. We pre-surveyed 53 humanities researchers, 93% of whom scored at the high scores threshold for each item in the DHQ. This indicates that humanities researchers in our sample generally tend to accumulate digital files. To ensure that the research sample is sufficiently representative and accurately reflects the overall diversity, thereby enhancing the credibility and applicability of the study, we adopted a stratified random sampling method. The 53 samples were divided into four strata based on gender, age, discipline, and education. Each stratum was then proportionally sampled at random. Ultimately, we selected 20 humanities researchers for interviews.

To alleviate potential biases caused by individual personality traits among researchers (Kumpulainen and Late, 2022), we employed triangulation. Three different researchers conducted interviews with 20 Chinese humanities researchers at different time periods (approximately weekly). A fifth of these interviews were conducted face-to-face in the first and second rounds, while the remaining were conducted remotely in the third round. Participants consisted of eleven university faculty members, five researchers from cultural heritage and memory institutions, and four current Ph.D. students. Of the participants, 9 were men and 11 were women, with ages ranging from 30 to 62 years old. They represented diverse humanistic backgrounds, encompassing literature, history, art, and other related disciplines. In order to protect the anonymity of the participants, we did not link specific details to individual participants (see Table 1). Fifteen of these interviews were conducted face-to-face, while five were conducted remotely. The duration of the interviews varied depending on the depth of the responses, ranging from 30 to 80 min. Participants were compensated $7 for completing the interviews (although some scholars participated out of interest and declined compensation).

Our interview outline was structured around the research questions and supplemented with relevant additional inquiries. The interviews commenced by gathering background

Characteristic	Category	Count	Characteristic	Profile category	Count
Gender	Male	9	Disciplines	Literature	3
	Female	11		History	4
Age	30~35	3		Philosophy	2
	36~50	14		Classics	1
	>50	3		Linguistics	2
	Postdoctoral research	2		Anthropology	2
Education	Doctorate/Ph.D.	13		Art History	3
	Master's degree	5		Archaeology	3

Source(s): Table by authors

Table 1. Overview of respondents' demographic background

information such as educational background, academic disciplines, and research areas. Subsequently, participants were asked to discuss their most frequently used tools and platforms for collecting digital content related to academic research, approximate categorizations of these contents, and their strategies for managing them. Additionally, the reasons behind their digital hoarding behavior were also explored at a general level. The conversation then shifted towards specific interview themes. We employed a variation of the Critical Incident Technique (CIT) (Flanagan, 1954), in which participants were asked to describe their typical experiences in digital information practices. By sharing their extensive engagement in digital information practices—conditions under which individuals are more prone to excessively consume digital content—we aimed to gradually explore the development of digital hoarding behavior. The CIT method assists interviewees in understanding behaviors and experiences in specific situations, such as a humanities research project, and has been widely utilized in information behavior research (Urquhart *et al.*, 2003).

Furthermore, with the respondents' consent, guided tours of their digital spaces were conducted to examine the phenomenon of hoarding, during which hoarded content was observed. Guided tours refer to participants leading researchers through the physical or digital information space under discussion and then providing a detailed tour of each (Vitale *et al.*, 2018). Ten of the guided tours were carried out face-to-face, while five were conducted remotely via screensharing; the remaining five were not presented due to personal reasons. During face-to-face interviews, the interviewee's hoarding situation was documented and photographed. In online interviews, participants shared their screens or provided screenshots for presentation. Consequently, we gained an accurate and authentic understanding of the interviewees' hoarding practices and engaged in more specific and targeted discussions during the interview process.

3.2 Analytical procedure

Each interview was transcribed into the MAXQDA qualitative coding software for encoding analysis. We employed thematic analysis (Braun and Clarke, 2006) to identify and extract categories related to digital hoarding behavior. This study utilized a two-stage thematic analysis approach (Cai *et al.*, 2020). Initially, we took a deductive approach to develop codes and main themes that captured the characteristics of hoarding behavior in the information space of humanities researchers, along with the underlying factors contributing to digital hoarding behavior. During this phase, based on the theoretical framework of existing literature and with the help of interview questions related to research questions, a start list of exploration topics is obtained, which is comprised of categories for the coding process (Azungah, 2018), as illustrated in Table 2. While applying this predefined encoding scheme, we also remained open to the possibility of new codes emerging beyond the predefined encoding scheme. The coding results were cross-checked among the authors to ensure consistency. Subsequently, adopting an inductive approach, we thoroughly examined the interview data to identify patterns and associations in humanities researchers' discussions of their perceptions, experiences, and coping strategies related to digital hoarding behavior. During this stage, the three researchers engaged in continuous discussions and revisions of the coding, condensing previous codes into overarching themes (Corbin and Strauss, 2015). Finally, we reached data saturation when scholars' narratives of digital hoarding behavior began to significantly overlap, and no new conceptual categories emerged.

4. Results

The digital scholarship age in the humanities has witnessed a rise in digital hoarding behavior. To gain a deeper contextual understanding of digital hoarding in scholars' research

Predefined category sources	Selected sub-questions based on interview questions	Corresponding RQ	Category (start list)
An a-priori model for overarching concept of 'digital hoarding' is derived by employing the three subconstructs termed as constant acquisition, discarding difficulty, and clutter propensity (Sedera *et al.*, 2022)	Do you feel in control of the digital content you keep? Do you retain digital content you might not use in the future? How do you curate and organize the digital content in your academic work? Do you maintain regular practices for sorting it? Are there any concerns you have about keeping the current amount of digital content you own?	RQ1	Excessively acquiring and reluctant to delete Poorly organizing
	How do you understand digital hoarding in your research practices?		New codes emerging beyond the predefined encoding scheme
Humanities scholars are influenced by historic, disciplinary work practices, by personal preferences, and by social, cultural, and environmental contexts affecting research practice (Given and Willson, p. 807)	Does the digital humanities paradigm affect your digital hoarding practices? How? Have the affordances and constraints of digital technologies or tools impacted your digital resource accumulation and deletion practices? How? Are your personal characteristics and habits related to digital hoarding? Why and how?	RQ2	Paradigm shift in the humanities Influence of personality traits Digital affordances
	Can you provide insight into why you engage in digital hoarding within your digital scholarship practices?		New codes emerging beyond the predefined encoding scheme

Source(s): Table by authors

Table 2. The category start list for characterizes and influencing factors (deductive coding)'

practices, we have elaborated on three thematic discoveries regarding digital hoarding behavior. Firstly, characteristics of digital hoarding among humanities researchers in their academic work have been identified. This involves synthesizing information management practices from various participants, both in terms of individual traits and commonalities, summarizing them to reveal overall trends and features of digital hoarding behavior. The second theme is related to influencing factors. Various factors contributing to this phenomenon were identified at different levels. The third theme involves humanities researchers' perspectives on digital hoarding and how they cope with it. The following sections discuss these themes and subthemes.

4.1 Theme 1: characteristics of digital hoarding in humanities researchers' daily work

As a form of scholarly work, digital scholarship in humanities involves the use of digital resources and technologies to support research and teaching. In this research context, we identified the characteristics of digital hoarding behavior, which manifest within the realm of

digital scholarly work and are reflected at three levels. These characteristics are summarized, along with example codes, in Table 3.

Excessively acquiring and reluctant to delete in scholarly work practices—In this study, humanities researchers frequently mention productive/retention actions toward digital

Characteristics	Specific manifestations	Examples
Excessively acquiring and reluctant to delete in scholarly work practices	Acquiring digital content itself is, to some extent, an "enjoyable" activity	Collect, document, store, keep, create, back-up, reproduce digital photographic, scan of artworks (Frequently mentioned retention actions)
	Indiscriminate storage of materials, information, or files	Millions of digital reproductions of physical artworks with little context or metadata, tens of thousands of databases of cultural artifacts, lots of personal digital diaries and correspondence, over 1,900g of collections of literary works
	Becoming not as eager to delete	Enhanced digital storage tools including generic tools (e.g. Dropbox, EndNote) and explicitly designed for humanities researchers (e.g. Omeka, https://omeka.org/); increased space expandability including personal and institutional devices
Poorly organizing and lacking motivation	Not carefully stored	Creating temporary backup folders haphazardly, casually storing downloaded files, piling files on the desktop, favorites feature being widely used for either social media or other platforms/tools quick access to digital materials
	Digital materials are roughly categorized and indexed	Date, occasion/event, and place are used to create personal categorization strategies, organizing is a vexatious activity
	Naming issues for digitized materials and cognitive burden due to context loss	Randomly naming, files named by devices independent from the context, losing the context they named files over the years
Difficulty in re-finding	Serendipitous discovery of hoarded materials	Encounter with hoarded content during browsing, navigate in day-to-day working
	Functions of tools failing humanities scholar on searches	Search features fall short in capturing the intricate nuances of scholars' memory and its associations; documentary, or visual-and audio-recordings searches not being advanced enough
	Memory loss erases hoarded information over time	Unable to locate, forgotten, humanities researchers favoring witnessing stockpiling with their memories
	Accumulate but do not process and share	Mass accumulation of knowledge that is preserved in knowledge-intensive workplaces and may or may not be shared with others in the future

Source(s): Table by authors

Table 3. Characteristics of digital hoarding among humanities researchers (N = 20)

materials in their academic work (except for linguistic researchers who use "data," while others prefer "research materials" or "primary sources"),such as document, collect, keep, create, back-up, reproduce scan. For most participants, acquiring digital content itself is an "enjoyable" activity. Three-quarters of the interviewees reported indiscriminately storing large amounts of overwhelming digital possessions/materials/recordings, mostly in the form of web-based, self-created or received. Specifically, P5 described her document practices with phrases like "mountains of data" and "piles of things." P11 mentioned his storage, "millions of digital reproductions of physical artworks with little context or metadata" and described his cloud services and institutional material archives as "dumpsters". Additionally, two-thirds of our interviewees further emphasize that the unlimited space on personal and institutional devices facilitates the storage of digital materials in their work environments, eliminating or deleting seems not very important. For example, a young linguistics and humanities researcher, in particular, mentioned, "*The convenience of carrying large amounts of data with these tools, without considering for physical space constraints, makes me not as eager to delete*. (P6)"

Poorly organizing and lacking motivation—Humanities researchers in our study most commonly perceived their folders, favorites, or computer desktops not as the best organizational practice but rather as a randomly stored mess. Their organized practices are generally straightforward, involving random downloads even directly piling files onto their desktops and saving them on temporary folders, rough folder hierarchies, and distinctive file naming. Three participants were found to regularly record and store digital material haphazardly in the course of their academic research. This was usually reflected in the haphazard creation of temporary backup folders, the random storage of downloaded files, and the stacking of files on the desktop. Most humanities researchers report adopting some categorization strategies; however, overall, the highest frequency is primarily centered around roughly categorizing folder hierarchies. For example, half participants mentioned, "*I've got folders inside folders, and then more folders tucked away. It's kind of like a digital jungle. Every single top-level folder is a chaotic digital archaeological site*! (P18)" This practice also involves naming digital materials, including the use of nomenclature such as date, context, relevance, or location. However, this approach creates a cognitive load for humanities researchers who indicate that they use materials by memorizing the naming context. According to P20, the volume of notes quickly became overwhelming, and the names of the materials he had labeled were blurred, "*I've named academic materials from various sources in a way only I understand. It's like a secret code that seemed logical at the time. Sometimes, I just try to piece the contexts behind these files*. (P7)"

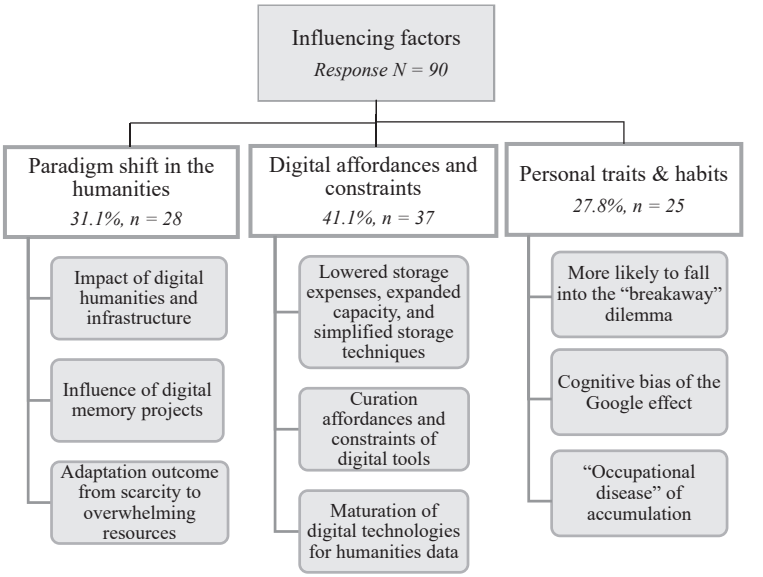
Difficulty in re-finding—"find it later" is an ongoing habit of postponing the use of digital content, but is often difficult to find and utilize because of the subsequent letting go. When participants were asked whether they used search or folders to find/return to information, almost all of them indicated that they engaged in re-finding, as it provided them with a false sense of control over their digital material, although there may be some difficulties. Four respondents mentioned a preference for the serendipitous inspiration that accumulated information provides, rather than intentional searching. As P6, an elder art history researcher, in particular, mentioned, "*Occasionally, it is the serendipitous discoveries that add a nuanced dimension to our research endeavors*." Navigating or browsing are common for encounters mentioned in this study, which are often constrained by specific contexts, typically resulting in lower occurrence rates. It also can be seen that allowing digital materials to become a hoard will make it harder to be efficiently located when needed. Some participants expressed that their memory loss, caused by saving too much or too varied information, erases hoarded digital content over time. A few participants also discussed the functions of existing storage tools to some extent failing them in searches. For example, P18, an archeology Ph.D. student specifically pointed out, "*Some of the tools or platforms I use are*

not transparent enough with extensive documentation for me to have a clear understanding of what I specifically own, aside from offering functions like scanning images using a thumbnail view." —reminiscent of what is commonly referred to in the corporate realm as 'dark data.' Unexpectedly, a few humanities researchers we interviewed hoarded digital materials related to knowledge from the network or other sources but chose not to share them with others. As one humanities researcher stated, "*I have gathered and retained a diverse range of digitized knowledge related to cultural landmarks, including from others . . . Just for my own amusement and may not be shared with others in the future*. (P7)"

4.2 Theme 2: influencing factors on digital hoarding behavior among humanities researchers

Several recurring themes emerged during the coding and thematic analysis process, providing insight into the main influencing factors and their intricate manifestations that impact digital hoarding behavior among humanities researchers. These factors can be divided into micro-, meso-, and macro-levels, corresponding to paradigm shifts, digital affordances and constraints, and humanities researchers' personality traits and academic habits, respectively. This classification aligns with prior research findings that highlighted similar influences on research practices, such as technology, disciplinary work practices, personal preferences, and social, cultural, and environmental contexts (Given and Willson, 2018). We found, in all, 90 responses of influencing factors observed from the interview data. Figure 1 illustrates our categorization of these factors based on their respective associated contexts.

4.2.1 Paradigm shift in the humanities. Individuals' information behavior is driven by response to environmental change or system change (McKellar *et al.*, 2023). The paradigm shift in the humanities has subtly influenced humanities researchers' research environment and approach (Kitchin, 2014). Most of our participants in this study reported that their information behavior is to some extent influenced by the paradigm of their own disciplinary



Source(s): Figure by authors

Figure 1.
The main influencing factors and corresponding manifestations leading to digital hoarding behavior

research. There is a significant possibility that this influence may give rise to a series of irrational information behaviors, such as digital hoarding.

4.2.1.1 Impact of digital humanities and infrastructure. Paradigm shifts in humanities research, such as the rise of digital humanities (Kitchin, 2014), have impacted the digital hoarding behavior of humanities researchers, i.e. the motivation of humanities researchers to hoard more digital content relevant to their scholarship in response to the changing academic environment. "*As a new paradigm in humanities research, digital humanities has significantly changed the way we work . . . like documenting, transcribing and annotating . . . one of the products is digital hoarding*. (P17)" Furthermore, the development and emergence of digital humanities infrastructure provide seamless access to information for humanities researchers. Under such conditions, scholars are more likely to 'binge' with digital content. For example, scholars familiar with humanities infrastructure said, "*The tools and resources in digital humanities infrastructure make it easier and more efficient for me to access, store and analyze. However, I must admit that . . . put more emphasis on pulling in and saving digital info or knowledge rather than contributing my own expertise*. (P9)" This response made us realize the prevalent phenomenon of knowledge hoarding in organizational work settings (Su, 2020). Subsequently, we asked other respondents (indirectly inquiring through knowledge retrieval and knowledge sharing, as knowledge hiding or hoarding is generally perceived as socially undesirable). Four relatively younger respondents indicated that they indulge more in accumulating work-related knowledge on the digital knowledge repository than in sharing or contributing "what they know. (P20, P17, P12, P3)"

4.2.1.2 Influence of digital memory projects. We found that one origin of digital hoarding behavior is related to the rise of digital memory in humanities scholarship, according to some respondents in history, archeology, and anthropology. With the integration of digital humanities and document archive management, digital memory projects (e.g. historical event memory and cultural heritage memory) that prioritize preservation and recording have thrived, attracting humanities researchers actively involved in keeping everything for projects (Gao *et al.*, 2022). It comes with both its upsides and downsides. Scholars are more susceptible to losing the balance between memory and forgetting due to their emotional attachment of memory to digital resources, leading to an accumulation of large amounts of digital content. "*Digital memory emphasizes recording and preserving digital data of our life events, I think it is meaningful for both individuals and collectives. So 'saving' is my default option . . . actually, what I am faced with is what to delete later*. (P11)" Another art history researcher discussed digital memory institutions, stating, "*Some institutions have committed to preserving data either for the project's duration, in perpetuity, or permanently. However, my colleague working in such institutions has raised concerns about the lack of a robust data management strategy and regular reviews, which may pose potential issues*. (P8)"

4.2.1.3 Adaptation outcome from scarcity to overwhelming resources. For many humanities researchers in this study, from an era in which information was relatively scarce to one in which information resources are overwhelming in size, access, and complexity (e.g. infobesity, information overload, and information anxiety) is often difficult. For instance, a scholar remarked meaningfully that he and his colleagues have discussed this issue, and they collectively believe, "*Digital resources for traditional humanities researchers are relatively scarce and difficult to obtain, requiring time and effort to search for. However, there are now multiple sources available for humanities scholars to obtain digital content, and many of them can be created or acquired with minimal effort and cost*. (P7)" A literature researcher expressed being overwhelmed with the quantity, "*As we navigate through our digital humanities journey, digitized content comes at us overwhelmingly. I am not ready to figure out how to record them yet, so I have no choice but to initially 'dump' them onto desktops, platforms, or some tools . . ..I have a feeling of being overwhelmed by increasing amounts of information.* (P3)" As such, maladaptive beliefs related to digital possessions among

humanities researchers, stemming from the availability of overwhelming resources, can also play a central role in digital hoarding.

4.2.2 Digital affordances and constraints. From our interview, we found that some affordances and constraints of digital technologies or tools appear to have a juxtaposed effect on the way digital content is documented, simultaneously facilitating and hindering. Participants almost universally tend to hoard digital materials because of their perceptions of digital technologies or tools that contribute to the mass undifferentiated retention of digital content and their experience of inaction in handling digital materials when using them. This inspires us to explore the formation and development of digital hoarding behaviors among humanities researchers in their digital scholarly work from the perspective of how particular systems and digital affordance affect user behaviors and practices—i.e. a digital materiality theory perspective (Kirschenbaum *et al.*, 2009; Nelson *et al.*, 2017).

4.2.2.1 Lowered storage expenses, expanded capacity, and simplified storage techniques. The increase in storage space (cloud technologies advancing), the ease of storage methods (one-click saving), and the abundance of storage tools used or used adaptively (Omeka, Dropbox, and Evernote) have all effectively challenged scholars to maintain control, contributing greatly to the digital hoarding behavior of humanities researchers. Unlike in the past, when digital data was stored on physical media such as CD-ROM and magnetic tape, humanities researchers can now easily generate large amounts of digital data in cloud space, various devices, or servers, and store it at a low cost. Technical tools provide unlimited storage possibilities for it. Some interviewees mentioned, "*I already have a massive library of materials on Evernote that I've accumulated over the last decade or so, the function of importing multiple sources files and sharing on it is convenient . . . although I know they're getting messy, I'm getting used to and relying on this tool.* (P15)" "*Sometimes I don't want to hoard, but Tropy is so tempting and convenient for recording images of artwork . . . especially content remains forever as time passes*. (P4)" Similarly, P12 described creating and storing video, audio, and multi-page documents through the Omeka, "*Omeka produces and stores thousands of files, and what I'm to do is remove those*. (P12)"

4.2.2.2 Maturation of digital technologies for humanities data. Several history and archeology researchers have mentioned that the digitization technology of humanistic resources provides a range of opportunities that enable scholars to easily accumulate and possess a large amount of heterogeneous and multimodal digital resources without necessarily "absorbing" them through close reading and note-taking. For example, imaging scanning technology allows for the digitization of paper-based materials, while 3D laser scanning and structured light scanning technology enable the digitization of cultural relics. P6 talked about his observations regarding this matter, "*Nowadays, whenever I visit archives or libraries, it's pretty much all about scanning and taking photos with my phone or camera (similar to how we used to use pens for handwritten notes), easily storing large amounts of content in palm-sized devices*. (P19)" It is evident that the research process for some humanities researchers has shifted from extensive reading and note-taking to scanning and gathering. In this sense, digital technology facilitates and allows scholars to accumulate as much digital information as possible, with the filtering and analysis of information typically occurring separately from the preserved moment or at a later time (Trace and Karadkar, 2017). This disconnect between extensive storage and utilization is likely to lead to digital hoarding.

4.2.2.3 Curation affordances and constraints of digital tools. The affordances of digital tools that allow for easy creation, storage, duplication, sharing, and long-lasting digital content, while enhancing the research practices of humanities researchers, can also provide a vehicle for digital hoarding, thereby breeding and exacerbating digital hoarding behavior among scholars. As mentioned by the participants, literature management software like Zotero can import a large amount of academic literature in bulk and can be interconnected and shared with other devices. Cloud storage service platforms such as Dropbox also require

only one click to download and store resources. Some historians reported how technology tools constrain their specific curated behaviors, indicating that existing digital tools lack sufficient curation functions (e.g. indexing, marking, correlation, and annotation). They also noted that the invisibility (the boundaries of digital space are not as readily observable to the user as those of physical space), poor searchability, and non-associability of digital content on digital tools could also limit their reuse and organization of digital content. As a result, these tools are more likely to cause users problems such as excessive preservation, high management costs, and difficulty in retrieving digital content, which also places higher demands on humanities researchers' personal information management in digital scholarly practice in the future. As one of our interviewees remarked, "*. . . many digital tools out there are great for saving, adding stuff, or just dumping things with a single click. But here's the kicker – sorting, adding metadata, or even just labeling stuff isn't as easy-peasy. What I've accumulated over time feels like this mysterious black box to me. I hope metadata should be created for storage, access, and analysis, of course, citation management*. (P1)" An archeology researcher raised the same concerns, "*When the bookshelves or room is filled, I could observe the increasing volume as it filled up, while also noting the available space . . . and I have my own 'best practice'. However, in a digital setting, these variations are not as easily noticed.* (P11)"

4.2.3 Influence of personality traits and academic habits. Our interview also revealed that if the digital affordance of digital tools allows humanities researchers to hoard digital content without realizing it, then their personality traits and academic habits will influence them to decisively and proactively delete redundant, low-value density, and weakly relevant digital content. Overall, three personal-related factors were identified:

4.2.3.1 More likely to fall into the "*breakaway*" dilemma. Whether the personality traits of humanities researchers are more likely to fall into the "breakaway" dilemma has yet to be empirically tested at the psychological level. Our results theoretically confirmed this. For example, one interviewee highlighted, "*I think I am more spiritually oriented . . . spend a lot of time reading and hoarding books that have nothing to do with my profession but are directly related to my material and spiritual life . . . Acting as a piler, I'm resistant to getting rid of these digital possessions . . . too difficult to determine what is to discard . . . even though I probably don't have time to use them*. (P20)"

Therefore, the digital hoarding behavior among scholars is deeply influenced by their personality traits. Furthermore, it seems like both senior and junior scholars are experiencing a fluctuating and struggling process of digital hoarding. Senior scholars tend to accumulate materials in a loosely organized manner and exhibit greater levels of digital disorganization than more junior scholars (Uğur and Çalışkan, 2022), perhaps due to their conservative personality, while junior scholars argue that they deal more with digital tools and resources than senior scholars, but it is also because of this that they get lost in the digital wave.

4.2.3.2 Cognitive biases of the Google effect. The Google effect refers to the phenomenon where people tend to rely less on their memory when they know that information can be easily obtained through the internet (Sparrow *et al.*, 2011). The acquisition and storage of digital information have replaced the previous process of individuals internalizing information. In our research, we found that humanities researchers are no exception to this trend. When describing their research processes, both P7 and P12 indicated that in academic research, they may excessively rely on online databases or other electronic devices instead of their own memory to recall literature. Their memory patterns and habits are slowly becoming alienated due to increasingly advanced digital storage methods. The digitization process, which some argue makes researchers "lazy," emphasizes the importance of obtaining and possessing digital information. "*When I genuinely needed these materials, I couldn't remember where I had stored them or the specific content. I simply recalled saving them with just one click and then continued to search and save again.* (P7)"

4.2.3.3 "*Occupational disease"* of accumulation. Unlike other groups (e.g. college students, internet users, and employees), humanities researchers' tendency to accumulate digital data and their reluctance to delete it often has deeper epistemological roots. Humanities researchers have always attached importance to document collection and accumulation, and this "occupational disease" has become even more "aggravated" in the digital age, i.e. many humanities researchers tend to hoard a lot of digital content they find useful or subconsciously think they may use in the future, as long as storage space allows. They often show hesitation about deleting or discarding contents in the curatorial process. Some respondents thus point out, "*Accumulating and expanding my collections seem like a priority for us humanities researchers, I've always been like this . . . This may be a common problem for us . . . In terms of the paper stuff, it is fairly random, and I frequently shove paper-based materials in different storage places such as shelves in the form of piles, files, piles of files, So it doesn't surprise me that this hoarding habit occurs in a digital environment*. (P8)"

4.3 Theme 3: humanities researchers' perception and response to digital hoarding behavior

Our third theme describes humanities researchers' subjective experiences and responses to the emergence and development of digital hoarding behaviors. Participants across disciplines shared a common nuanced feeling: that hoarding is perceived as both positive and negative. The positive aspect is associated with their expectations, while the negative aspect is intertwined with their experiential perceptions. A noticeable gap exists between these two perspectives.

First, we identified several positive aspects of hoarding behavior based on their expectations. Hoarding is perceived as an artistic and aesthetic pursuit for some art researchers in this study who take pleasure in creating and preserving self-generated original artifacts, such as consistently archiving documentary audio and video and digitally replicating physical artifacts. According to one scholar, "*More is better . . . art, artifacts, or other aesthetically valuable objects are cherished elements of my creativity and aesthetic experience . . . pleasing to look at . . .*". Impressively, several researchers appreciate the vast and disorganization because it sometimes offers them serendipity and inspiration from their digital knowledge repositories. For instance, one art researcher explicitly expressed a dislike for purposeful and deliberate information searches in personal repositories (hoarding), preferring instead to casually browse. He finds inspiration through random exploration, relishing the surprise and unexpected discoveries that arise while stumbling upon information retained as part of his research work. As a result, he tends to retain digital information and refrain from deleting it, even when it doesn't appear immediately useful. Additionally, another interviewee, with some hesitation, suggested that "*ideally, digital hoarding implies providing a sense of uncertainty avoidance . . . at least on a psychological level and it is easy and not time-consuming.* (P18)" echoing Dillon's (2019) notion that it serves as a defense against negative emotions and a desire for control. Similarly, a researcher working at cultural heritage and memory institutions stated that hoarding serves as a coping mechanism for uncertainty at work, saying, "*I can't bear the thought of suddenly needing a specific document at work that I don't have, even in the archives or as a backup*". In these ways, humanities researchers tend to be somewhat more accepting of hoarding, distinguishing themselves from other demographic groups.

However, some participants recognized that the hoarding of information was problematic, yet found it difficult to delete anything. The potential issues of digital hoarding affect both their academic practices objectively and induce cognitive distress subjectively. Several researchers mentioned that digital hoarding led to problems such as distraction and reduced work efficiency (P2, P7, P14), hindering researchers from focusing on their tasks. One participant expressed, "*It's really hard to keep those research materials of mine labeled and*

organized all the time, whether work-based or research-based. The best I can do is have a unified folder. I actually expected it to be more efficient . . . it always distracts me from what is important . . . (P2)" Moreover, information fragmentation occurred across devices, platforms, project spaces, formats, and versions, affecting participants to varying degrees and triggering negative emotional responses. Some even experienced negative emotions such as information overload, frustration, and exhaustion as hoarding became more severe (P3, P9, P12). Three other respondents, discussing the infusion of digital material they hoarded in knowledge management scholarship, talked about knowledge hoarding in casual conversations as a potential impediment to the development of digital scholarship. For example, P9 mentioned, "*It also seems to be an accumulation of knowledge, and perhaps the only problem is that I am deeply enthusiastic about acquiring and retaining rather than sharing my work-related knowledge . . .*"

Negative experiences (emotions) associated with behavior may also suggest a gap between actual and ideal behavior (Alon and Nachmias, 2020a). In our interviews, one participant casually mentioned, "*I idealistically perceived digital hoarding as relatively harmless as long as it remained undisturbed*". In his view, a balance between digital hoarding and regular collection behavior is achieved only when undisturbed and unaffected. To bridge this gap, some respondents in our interviews also employed coping strategies in response to the stress caused by hoarding. The reasons and effects for the respondents' adoption of these two coping behaviors are given in Table 4.

From humanities researchers sharing their perceptions and experiences regarding digital hoarding, two behaviors were observed in response to bridging the gaps of digital hoarding as best as possible: decluttering and avoidance. Some respondents facing the threat of digital hoarding, such as memory warnings, stress, and anxiety, tend to choose actions such as deletion, categorization roughly, or digital disconnection (Nassen *et al.*, 2023) as coping mechanisms. However, under the influence of various factors, hoarding behaviors may rebound (P8). For instance, an experienced researcher studying the history of arts shared the negative experiences that he had with digital hoarding:

> . . . At times, I unexpectedly come across a large pile of digital materials while using tools like Omeka or Zotero. It can happen by accident or when I'm reminded about limited storage space. In those moments, I feel confused and frustrated, as if I no longer have control over these amassed digital assets. This usually leads me to take passive steps, like organizing, deleting, or adding tags to restore some sense of order. . . . However, 'ad hoc storage' is pretty handy . . . Before I realize it then, the issue of digital hoarding pops up again. (P8)

Additionally, some scholars exhibit avoidance behaviors as a strategic action for emotion regulation, such as no response and "letting it go", procrastinating to deal with, and circumventing deletion by resorting to "additional digital spaces", rather than adopting effective digital curation strategies. However, this strategy can only help individuals temporarily alleviate negative emotions and cannot change the actual stress environment. This is despite the mandate for individuals or organizations in the humanities to excel in digital curation, including creating metadata, devising digital storage strategies, and guaranteeing the longevity and accessibility of digital assets to support their research work (Poole, 2017). Interviewees' attitudes toward the digital curation requirement appear to be hesitant, and in some cases, even neglectful. Several participants acknowledged the problematic nature of such loose digital information management practices which contradicted the effective digital curation advocated within their academic disciplines (P4, P19). A scholar working in the field of linguistics discussed her experience of non-response to this behavior:

> Yes—in general, my field (linguistics) tends to generate loads of corpus data, some of which are useful, while others are just plain unnecessary. I really don't like spending my time and energy on

Coping behavior	Reasons	Representative quote	Effects
Declutter deleting categorizing roughly digital disconnection	Reduce academic efficiency	"I frequently lose track of those materials, leading me to invest time in repeating the same search process, So, I force myself to name each file more complex and unique, which may be better for finding."(P17)	Temporary alleviation of digital hoarding, with no real management applied to it
	Perceived stress and anxiety	"In severe situations I see my cluttered and disorganized folders, it makes me feel quite exhausted. [. . .] Sometimes, I even abandon or stop[overloaded USB/application] to improve this messy situation."(P9)	
	Limited memory	"[. . .] This doesn't happen as often as before because storage costs have decreased, but it still happens from time to time. You hardly notice it until you get a warning that your digital storage is almost full[. . .] some delete actions should be taken"(P3)	
Avoidance no response no response relocating	Avoid negative emotions	"[. . .] a way to temporarily escape the confusion or pressure brought about by the overwhelming digital content – to pretend I haven't hoarded them and turn a blind eye to them."(P20)	Aggravating the development of digital hoarding
	A waste of time	"Because yes, I don't want to spend time on these things . . . tidy up this massive and chaotic "digital library" maybe take a whole lot of effort" (P11)	
	Low self-control ability	"[. . .] to tackle this problem myself, especially by using some effective techniques or tools, feels like a real challenge to my abilities. So, I give up [. . .] and let it go."(P9)	
	Inspiration by chance	"Serendipity is an amazing experience [. . .] Sometimes, inspiration just comes out of nowhere, like from a jumble of random stuff. It's more memorable than purposefully searching for things"(P14)	

Source(s): Table by authors

Table 4. Reasons and effects for coping behavior in response to digital hoarding behavior

dealing with material that's particularly useless; even a quick glance at it can make me feel extremely uncomfortable, and it gives me a headache . . . Just steer clear of it!. (P4)

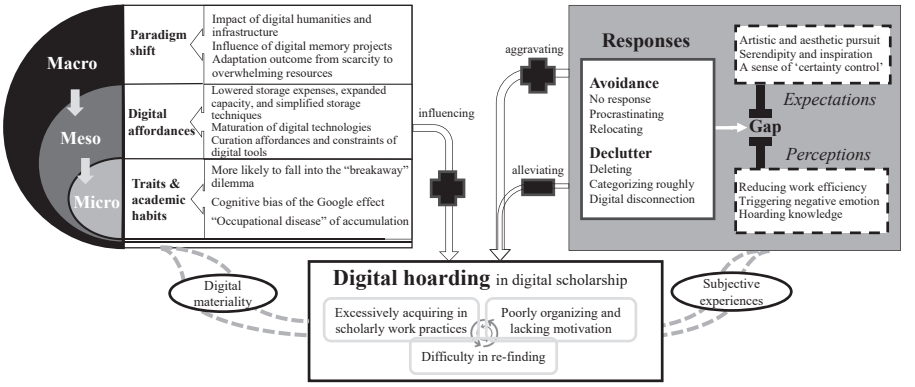
5. Discussion

We have observed that the characteristics of digital hoarding behavior among humanities researchers in digital academic work are manifested through the interconnection and interaction of several facets, including excessively acquiring and being reluctant to delete scholarly work, poorly organizing and lacking motivation, and having difficulty in re-finding information. In addition, digital hoarding behaviors in humanities researchers' digital

scholarly work environment are influenced by two forces. One is the various stimulating external environmental factors under the influence of digital scholarship; that is, digital hoarding behaviors are a long-cycle, complex behavioral practice co-evolved in multiple factors such as research paradigms, technologies, and humanities researchers individually. The other force is the thoughts, opinions, feelings, and behaviors (subjective experiences) that emerge from an individual's interaction with the digital environment. Incorporating a broad, connected, and socially contextualized perspective on a humanities researcher's academic information management behavior can give it deeper meaning (Conrad *et al.*, 2020). To comprehensively understand this emerging phenomenon, this research developed an integrated digital hoarding behavior storyline model of humanities researchers in the academic process (Figure 2) and presented several key findings to address the three research questions:

5.1 Digital hoarding can be viewed as a manifestation of a disconnect from expected or ideal PIM among humanities researchers

With regard to the RQ1 (i.e. what characterizes the digital hoarding behavior of humanities researchers in everyday digital scholarship?), the results of theme 1 revealed that humanities researchers exhibited three types of digital hoarding behavior characteristics that are highly nested. The first often-discussed characterization of digital hoarding behavior is excessively acquiring in scholarly work practices—As indicated by several responses, they have an overwhelming interest in collecting digital content—ebooks, manuscripts, and digital images—even if the information is minimally needed and used. Humanities researchers prefer engaging in productive activities related to PIM across multiple platforms in their workplace settings, such as scanning, reproducing, and documenting. They express a lack of bounded control over the accumulated content, often not knowing the location of certain information. While saving "everything" may not be a desirable goal in response to the explosion of information (Ray, 2012), it has been observed in workplaces, households, and personal contexts of hedonic social media engagement (Oravec, 2018). The second characteristic is poorly organizing and lacking motivation—Humanities researchers reported that their (hoarders') digital storage spaces often appear disorganized and compromised due to the generation and storage of these data spaces being relatively hidden (Drosopoulou and Cox, 2020)—at least for those using cloud storage or services (Thylstrup, 2019).



Source(s): Figure by authors

Figure 2. An integrated model of digital hoarding behavior in digital scholarship among humanities researchers

The third characteristic is difficulty in re-finding—simply accumulating without using. Even though many studies emphasize that finding relevant information at the point of need is essential for it to be used rather than stored away and forgotten (Pontis *et al.*, 2017). Previous research indicates that academics in an iSchool place little value on their digital material beyond immediate use (Drosopoulou and Cox, 2020). A striking finding of our study suggests that although humanities researchers have difficulty realizing the value of their accumulated content through re-finding (see Table 2), they still hold an anticipated value for their hoarded digital materials. This includes, but is not limited to, collecting for future use, part of their creative process and personal enjoyment (Carr, 2024), affirming Drosopoulou and Cox's (2020) notion that many humanities scholars have a different attitude toward the value of their files due to their interest in the creative process.

The three-fold characterization we theorized of digital hoarding among humanities researchers is largely consistent with previous research on the conceptualization of digital hoarding (Neave *et al.*, 2019; Sedera *et al.*, 2022; Sedera and Lokuge, 2018). However, differences can also emerge due to variations in humanities disciplines' practices. Firstly, humanities researchers tend to hoard digital materials with a stronger continuity and larger memory, such as extensive archives of historical documents, vast collections of literary works, comprehensive databases of cultural artifacts, or even personal digital diaries and correspondence. This differs from the fragmented "like and collect" style of information hoarding on social media platforms among internet users, stimulated by recommendation algorithms (Wu *et al.*, 2023). Secondly, the characteristics of digital hoarding intersect with and extend the challenges of PIM, such as information overload, retrieval difficulties, and determining the future value of information (Chaudhry and Alajmi, 2022). Digital hoarding can be viewed as a manifestation of a disconnect from expected or ideal PIM practices among humanities researchers. From a group differences perspective, some academics in other disciplines place little value on digital possessions, with effective PIM limited to browsing information and rarely involving curation. For these individuals, digital hoarding is inconsequential and may not even be perceived as hoarding (as digital hoarding is more of a value-based perception, while PIM is an objective practice) (Drosopoulou and Cox, 2020). However, the dependency on digital literature, sporadic interest in the creative process, and relatively low information literacy among humanities researchers lead them to have a different attitude towards the value and management of their digital content. This also explains why the third characteristic we identified emerges from other existing definitions of digital hoarding, as it reflects the humanities researchers' desire to realize the value of their accumulated content.

5.2 Paradigm, digital affordance, and personal factors shaping digital hoarding behavior

Moving on to RQ2 (i.e. what factors have triggered digital hoarding behaviors among researchers in their digital scholarship practices?), digital hoarding is jointly affected by responses to paradigm shifts and technological opportunities in the digital scholarly practices of humanities researchers, although it ultimately depends on their traits and personal preferences. This confirms that problematic information behavior is deeply rooted in various stimulating external environments in times of changes at work, as well as the subjective experience of individuals in their interaction with the digital environment (Ruthven, 2022). In this sense, digital hoarding is a product of the digital humanities paradigm, technology, and individual traits, reflecting a common phenomenon within this specific group in digital scholarship. This suggests that humanities scholars' hoarding behavior is not solely driven by pathology (van Bennekom *et al.*, 2015). Tracing back to the PIM theories that are deeply intertwined with digital hoarding behavior proves equally fruitful. PIM is inherently personal but also heavily influenced by circumstances, making a

collective analysis of variables more illuminating (Dinneen and Frissen, 2020). Considering these findings as a whole suggests that personality does not stand out as a singular determinant of digital hoarding.

At the macro level, paradigm shifts (i.e. digital humanities and infrastructure, digital memory projects, and the shift from scarcity to an abundance of resources) can create fertile ground for the creation and accumulation of digital content and the digitization of scholarly work (Late and Kumpulainen, 2022). Simultaneously, these paradigm shifts accelerate the development of disciplinary technologies and tools, increasing humanities researchers' interaction with digital resources and tools and providing "opportunities" for digital hoarding to breed. Digital materiality theory further helps us anchor that digital affordances and constraints are the primary influencing dimension driving humanities researchers toward digital hoarding. This theory states that the affordances and constraints of digital technology facilitate a better understanding of the impact of digitization on individual behavior (Kitzie, 2019).

Similarly, within the context of PIM, several studies have highlighted that tools and operating systems used for tasks can enable, restrict, and influence user behavior (Dinneen and Frissen, 2020). For instance, the file size limits imposed by different operating systems can prompt users to create deeper and more extensive folder structures (Dinneen and Frissen, 2020), while constraints such as limited disk or cloud storage space may dictate user storage practices (Kljun *et al.*, 2016; Dinneen *et al.*, 2019). These findings align with the digital affordances and constraints identified in our study Digital hoarding behavior therefore is somewhat closely intertwined with PIM activities, albeit with varying degrees of detail. The factors we have identified—such as near-zero digital storage costs, easy storage methods, curation affordances and constraints of digital tools, and maturity of digital technologies for handling humanities data—have both refined and expanded the development of PIM systems. These factors also provide a foundational understanding of how technological influences contribute to problematic information behaviors, including digital hoarding. Furthermore, from a digital affordances and constraints perspective, we have found that there is preoccupied with productive actions involving data and information, such as creating, storing, accessing, and (re)using while neglecting the complementary aspect of the information/data lifecycle: deletion. In fact, research has shown that accumulating files has little impact on hoarding behavior, whereas failing to delete files can make a difference (Tugtekin, 2022).

Many past studies conclude that the ways individuals store, manage, and utilize digital information are deeply personal (Hyldegard, 2009; Dinneen and Frissen, 2020). Our study has identified three new personality traits closely associated with humanities researchers: Humanities researchers are more likely to encounter the "breakaway" dilemma, triggering the "occupational disease" of accumulation in the digital era, and amplifying the cognitive bias of the Google effect. These factors are consistent with humanities researchers' historical academic habits (Al-Omar and Cox, 2013; Pontis *et al.*, 2017), and are tied to digital hoarding. While previous research has explored the influence of grouped traits such as the Big Five personality traits (Bergman *et al.*, 2020), emotions (Whittaker and Massey, 2020), and memory (Bergman *et al.*, 2020) on PIM, there is still limited understanding of how individual characteristics of specific groups influence their digital hoarding behaviors. Our study further identifies and explains specific personality traits directly related to humanities researchers' digital hoarding practices.

Surprisingly, some studies on digital hoarding have pointed out that individuals often tend to "anxiously" hoard vast amounts of useless personal data (Wang *et al.*, 2023). There is also literature that suggests digital possessions are considered a part of personal identity (Cushing, 2011). In our study, humanities researchers reported little anxiety- and personal identity-driven digital hoarding. This is because they generally felt that messy or even

useless digital repositories could not be extended into personal identity and construction. The survival of digital materials, they argued, lacked a sense of ritual compared to physical hoarding. This finding further confirms that the way humanities researchers approach the management of personal information will influence the value they assign to digital materials (Post, 2017). Additionally, considering the limited geographical scope (China) of the research sample collection, humanities researchers encounter the "breakaway" dilemma perhaps because they are influenced by the traditional Chinese mantra of "物尽其用" ("Avoid waste" in English), which advocates not discarding anything that might prove useful in the future. This finding echoes Timpano *et al.*'s finding that Chinese populations, including humanities researchers, exhibit a stronger inclination toward beliefs about saving compared to their counterparts in the United States (Timpano *et al.*, 2015).

In summary, these factors we identified, such as paradigm shifts in the humanities and individual traits and habits, are relatively unique to the context of digital scholarship, and Several technological aspects of digital hoarding are promising and often-discussed factors that partially overlap with existing research at PIM (Dinneen *et al.*, 2019; Dinneen and Julien, 2020), partially aligning with findings from previous studies, such as Sedera *et al.* (2022), although previous studies have rarely explored systematically in the context of digital scholarship for humanities researchers.

5.3 An experience of the overwhelmingly unmanageable yet inspiring digital hoarding by humanities researchers

RQ3 is addressed (i.e. How can humanities researchers perceive and cope with the negative impacts of digital hoarding behaviors in their digital scholarly practices?). Most hoarding experiences in humanities researchers' work were perceived by most participants as involving a duality of good and bad. They did not simply view hoarding as an inefficient information behavior that always needed solving or optimizing. Instead, they held seemingly contradictory attitudes and responses toward digital hoarding. We characterize this as an expectations-perceptions gap. On one hand, hoarding is a significant concern for humanities researchers, especially for arts and humanities researchers, who view it as an artistic and aesthetic pursuit. For them, accumulating digital artworks is a marvelous experience, even if they do not use them for purposes such as academic research and work, but simply to enjoy them occasionally. Furthermore, the most encouraging finding of this study is that humanities researchers view their hoards of personal information as potential inspirational material, not just objective evidence/data, including useless material that may not appear to be directly related to the field of study, confirming the findings of Erdelez and Makri (2020), Hellmich and Dinneen (2023). However, to the best of our knowledge, PIM practices or tools do not currently consider the role of information encountering in the context of digital hoarding on the scholarly work of humanities scholars (Reviglio, 2019). Moreover, some hoarders exhibit possessive tendencies, regarding their possessions as functional (Hulber, 2020). A small number of respondents to this paper understand hoarding behavior as seeking a sense of control over uncertainty, considered from a utilitarian perspective. Retaining everything may create a false sense of uncertainty avoidance (Neave *et al.*, 2019). This finding could speak to a cultural phenomenon, wherein Chinese populations, influenced by the traditional cultural notion of "有备无患" (being prepared for all eventualities), tend to exhibit higher levels of uncertainty avoidance than Americans (Hofstede, 1980).

On the other hand, the negative perceptions of hoarding faced by humanities researchers included two common challenges and one unique challenge. As discussed in current PIM literature regarding the increasing complexity of PIM and the associated challenges (Kumpulainen and Late, 2022; Zhao and Yang, 2019) and negative emotional experiences during the PIM process (Alon and Nachmias, 2020b), the humanities scholars' excessive,

unorganized, and ineffective preservation of digital materials can have a significant impact on their work and lives. Additionally, digital hoarding can lead to cognitive and emotional distress when it reaches a certain extent (van Bennekom *et al.*, 2015). Similar negative emotional issues have been documented in prior research on personal information management, where frustration and loss of control are common characteristics in interactions with personal information among the general population (Alon and Nachmias, 2020b). Humanities researchers are no exception, who often encounter many information projects in their scholarly work and are frustrated by not being able to process digital information items they have accumulated and hoarded. Our findings further emphasize the significant frustration resulting from discrepancies between expectations and actual perceptions. Last but not least, digital hoarding behaviors may be more likely to manifest at a higher level in workplace scenarios—knowledge hoarding (Su, 2020). The correlation between digital hoarding and knowledge hoarding in the workplace may be linked to the close relationship between digital scholarly work for humanities researchers and knowledge-intensive scholarly activities. It could also serve as one of the explanations for why information management and knowledge production have always been central concerns in humanities scholarship (Clement and Carter, 2017). The phenomenon of interviewees introducing knowledge hoarding when discussing digital hoarding also indirectly reflects the fact that digital scholarship is emerging as a new means for researchers to engage in knowledge production and exchange within digital environments. This finding extends and strengthens existing research on the dark side of organizational knowledge management related to knowledge hoarding (Su, 2020).

Humanities researchers involved in this study also exhibit certain responses when faced with this challenge and threat. Stress coping theory suggests that when individuals perceive stressful events or threats as changeable or controllable, they are more likely to adopt problem-focused positive coping strategies, such as planning and taking action (Folkman *et al.*, 1986). On the other hand, when the problem is uncontrollable or cannot be resolved in the short term, individuals tend to resort to emotion-focused negative coping strategies, such as avoidance, distancing, and forced displacement (Bala and Venkatesh, 2016). Although humanities researchers have taken certain actions to alleviate digital hoarding, such as deleting, categorizing roughly, and digital disconnection, it is apparent that they have not employed any advanced strategies to cope with these challenges. However, they also instinctively adopt avoidance measures, which seem to aggravate the formation and development of digital hoarding. These measures include not taking action in response to hoarded content or engaging in passive avoidance, such as transferring and rejecting. Prior studies have revealed a different role for negative emotions, arguing that they can prompt the creation of more folders with fewer files and deeper folder structures (Whittaker and Massey, 2020). Some participants in our study, however, have lost the motivation to participate in any PIM activities because they consider PIM activities to be a waste of time, even though they have employed idiosyncratic management practices in recording digital research practices (e.g. creating metadata, a form of organizing and managing) (Given and Willson, 2018). This could arise from significant disparities between individuals' actual and desired PIM behaviors, potentially undermining their motivation to manage personal information and eliciting negative emotions like frustration (Alon and Nachmias, 2020b).

6. Contribution and implication

6.1 Theoretical implications

First, this study proposed a conceptual model of digital hoarding behaviors in humanities researchers' digital scholarship. Through this model, scholars can gain better insight into how humanities researchers approach (or neglect) digital hoarding and the struggles therein,

as some humanities researchers perceive it as inspiring. This model may also be partially applicable to understanding digital hoarding in other groups.

Secondly, the theoretical framework of this paper integrates the perspectives of digital materiality and individual subjective experiences to enhance our theoretical understanding of the social, technological, and psycho-cognitive factors driving digital hoarding among humanities researchers within the context of digital scholarship across macro, meso, and micro levels. This integrated research perspective has the potential to inspire fresh insights among scholars regarding the causes of digital hoarding behaviors in humanities studies and other domains. Furthermore, it has broadened the scope and research boundaries of digital materiality theory within new research contexts.

Third, we offer nuanced insights for understanding digital hoarding among humanities researchers as a complex phenomenon or behavior, rather than simply as a problematic information behavior. Our theoretical framework conceptualized the dual experience of anticipated positives and encountered negatives as the "expectation-perception gap." From this perspective, digital hoarding can be framed as a fluctuating and challenging process rather than simply a direct problematic outcome. Therefore, this framing enables us to understand humanities researchers' digital hoarding behavior from both problematic and inclusive perspectives. When viewed through a problematic lens, we may consider issues like excessive accumulation of digital materials causing clutter and inefficiency. Conversely, from an inclusive standpoint, we can explore digital hoarding as a coping mechanism or a way to preserve valuable information and creative inspiration.

Fourth, unlike traditional research, which has focused on the most common digital activity and the most hoarded document type in the workplace—email hoarding (van Bennekom *et al.*, 2015; McKellar *et al.*, 2020)—digital hoarding behaviors are manifested in all aspects of digital activity across different work contexts and organizational cultural differences. In digital scholarly work contexts, humanities researchers hoard a wide range of digital content that is not limited to hoarding emails. This study explored multimodal digital hoarding (e.g. visual materials, original documents, ebooks, etc.) in digital scholarly contexts, providing new theoretical insights into digital hoarding in the new context of digital scholarship and extending our understanding of digital information management for humanities researchers.

Finally, this study will expand the traditional research field of digital literacy and personal information management (PIM) in humanities studies, offering new ideas and perspectives for promoting healthy and beneficial development. In traditional research on information behavior and personal information management of humanities scholars, researchers emphasize the importance of information seeking and collection but pay less attention to preserving and managing personal information (Stone, 1982; Drosopoulou and Cox, 2020). With the development of digital information technology, issues such as information overload and excessive storage have gained increasing attention. In our research, the threefold characterization of digital hoarding behavior and its potential consequences for humanities scholars provide another research perspective for optimizing and enhancing personal information management and digital literacy.

6.2 Practical implications

The research findings also offer practical insights for humanities researchers, digital scholarship service providers and institutions, and digital tool developers on how to alleviate digital hoarding in academic practices by adjusting both social and technological factors.

For humanities researchers and digital scholarship service providers or institutions, the research findings suggest that humanities researchers often respond to perceived powerlessness regarding hoarded content and behaviors with 'cold handling,' involving avoidance (i.e. no response, procrastination, and relocating), along with negative experiences

associated with digital hoarding. In light of these findings, the awareness and the practice of 'digital minimalism' (Vitale *et al.*, 2018) may emerge as a new form of digital literacy for humanities researchers in the future. Especially for humanities researchers who view digital hoarding as a dilemma, it is important to proactively limit the amount of humanities data, focus on curation and deletion, and avoid adding unnecessary cognitive loads and increasing distress. Given our study insights into the fact that some digital preservation institutions or organizations commit to permanent preservation but exhibit behaviors that avoid adopting standard digital curation practices, which can result in significant cumulative losses for the organization (section 4.2.1, paragraph 3), the adoption of institutional retention schedules should be promoted.

As per our unexpected finding that some respondents prefer to accumulate work-related knowledge rather than actively sharing it, sharing policies are likely to be as helpful as institutional retention schedules (Gormley and Gormley, 2012). Humanities researchers can share their accumulated content in order to trim down and activate individual personal information collections/space. Institutions (cultural heritage and memory institutions, universities, and other information institutions) must establish clear data management policies, distinguishing between what needs to be retained for organizational purposes and what is being hoarded. Emphasis should be placed on reviewing and organizing digital assets in the workplace, and policies should be developed to promote information sharing. Last but not least, the expectations of humanities researchers regarding digital hoarding—such as inspiration, aesthetic pursuit, and uncertainty avoidance—also indicate a theoretical perspective that does not overly impose the responsibility of minimalism and tidiness on humanities researchers, and does not excessively view their digital hoarding in academic work as a disease (Işliyen, 2019) or pathological behavior (Hulber, 2020). As Copic Pucihar *et al.* (2016) pointed out, chaos is adaptive and can only be controlled, not easily eliminated.

For digital tool developers, as Steinhart *et al.* (2012) suggest, a lack of oversight may lead researchers to store and back up their data in manners that do not optimize its potential for sharing and reuse. To maintain the best management practices and usefulness of digital materials, adequate metadata and ongoing management and oversight of digital resource storage tools can be developed by considering the insights gained from our study. Additionally, as digital hoarding among humanities researchers may be associated with invisibility, poor searchability, and non-associability of digital content on digital tools, Reminders, visualizations, transparency, and incentives for deletion in the design of digital tools are considered an effective means of getting people to subtract (Adams *et al.*, 2021). These measures aim to adjust and optimize scholars' digital material retention—quantity, types, and associations—by leveraging digital affordances and constraints. This, in turn, helps alleviate systemic problems related to hoarding in digital scholarly work for humanities researchers and improves the overall effectiveness of digital academic practices. Moreover, humanities researchers' positive expectations or beneficial perceptions of digital hoarding inspire us to offer them digital affordance for utilizing hoarded content, such as enabling one-click favorites and facilitating unexpected discoveries of hoarded content. However, to our knowledge, current PIM practices or tools do not consider the role of serendipity and inspiration amid digital hoarding in the academic work of humanities scholars (Reviglio, 2019). Therefore, options could be presented at the outset of saving to enable users to assess their reasons for saving specific digital content. If hoarding occurs, an intelligent algorithm could align the initial saving rationale with content relevant to the humanities researcher's current research, complemented by some randomized content presentation to enhance serendipitous information discovery (Hellmich and Dinneen, 2023).

7. Limitations and future research

The research findings of this study may be limited by geographical location. Despite the diverse disciplinary distribution of the respondents, all come from the same country. Therefore, it is necessary to interpret and apply the research results cautiously, as these results may not apply to all countries or regions. Future research can further validate the universality of the results through cross-national comparative studies.

The humanities encompass a wide range of academic disciplines. To enhance the generalizability and reliability of our research findings, we surveyed researchers from various specialized fields within the humanities. However, this approach inevitably overlooked certain specific research perspectives. In the future, it may be worthwhile to investigate the unique digital hoarding experiences of scholars in specific disciplines, such as historians, to uncover more specific and distinctive insights.

This study provides a preliminary exploration of digital hoarding behavior among humanities researchers, aiding our understanding of its formation and development. Future research can delve into the effects of digital hoarding by various humanities research teams on their output and project team cooperation, allowing for a quantitative assessment of its impact on the humanities community.

Due to the research methodology of semi-structured interviews, the majority of the findings were based on self-reports. Although this study partially addressed this issue by validating the information and actual performance of respondents as stated in their self-reports through the guided tour methodology, future research could incorporate multi-source responses to minimize the gap between self-reported behavior and actual behavior.

Besides humanities researchers, other groups such as organizational employees (McKellar *et al.*, 2020, 2023), college students (Alon *et al.*, 2020; Sillence *et al.*, 2023), and ordinary internet users (Sedera *et al.*, 2022) are increasingly engaged in digital hoarding. However, few studies have empirically explored the relationship between digital hoarding and digital well-being among these groups, as digital well-being is related to an individual's ability to effectively and healthily use digital technology (Parry *et al.*, 2023). Yet, it remains to be studied whether there are exceptions to this for humanities researchers, as digital hoarding can support unexpected discoveries and inspiration acquisition.

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Further reading

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Appendix

Table A1. Digital hoarding questionnaire

Dimensions	Measurement terms
Difficulty deleting	1. I find it extremely difficult to delete old or unused files
	2. Deleting certain files would be like deleting a loved one
	3. If I delete certain files I feel apprehensive about it afterward
	4. I strongly resist having to delete certain files
	5. Deleting certain files would be like losing part of myself
	6. Thinking about deleting certain files causes me some emotional discomfort
Accumulating	7. I feel strongly that some files might be useful one day
	8. I tend to accumulate digital files, even when they are not directly relevant to my job
	9. I lose track of how many digital files I possess
	10. At times I find it difficult to find certain files because I have so many

Source(s): Figure courtesy of Neave *et al.* (2019)

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